Mingyuan Zhang

Boston, MA Homepage zhang.mingyua@northeastern.edu 781 219 1785 Google Scholar

SUMMARY

Ph.D. candidate in Computer Engineering at Northeastern University, focused on **efficient foundation models** and **embodied AI**. Research spans parameter-efficient adaptation, sparse training, and mixture-of-experts for LLMs/VLMs, with industry experience in world-model-based robotic reasoning at Amazon and robust computer vision at Socure.

EDUCATION

***Northeastern University* (NEU)** Boston, MA
- **Ph.D.** candidate in **Computer Engineering** at SmileLab Sep. 2023 – May 2028 (expected)
- Advisor: Prof. Yun Raymond Fu
- **Research Interest:** Foundation Model | Efficiency | Embodied AI | World Model | Agents | Generative Model

***Northeastern University* (NEU)** Boston, MA
- **Master of Science** in **Electrical and Computer Engineering** Sep. 2021 – Dec. 2022
- **Concentration:** Computer Vision, Machine Learning and Algorithm

***Xidian University* (XDU)** Xi'an, China
- **Bachelor of Engineering** in **Telecommunications Engineering** Sep. 2017 – Jun. 2021
- **Honor of Excellent Student Leader**, 2020

EXPERIENCE

Northeastern University | **SmileLab** | *Research Assistant* | Boston, MA *Sep. 2023 - Present*

Advisor: Prof. Yun Raymond Fu

- **Foundation Models**
  - Proposed *Mask Fine-Tuning (MFT)*, a novel parameter-efficient adaptation paradigm that learns masks on LLM/VLM and improves performance through pathway selection.
  - Developed *MASKerade*, a novel MoE framework for LLM/VLM that learns different masks as experts while keeping pre-trained weights frozen.
- **Pruning and Distillation**
  - Proposed *Sharpness-Aware Random Sparse Training (SA-RST)*, enabling effective training of randomly initialized sparse neural networks without dense pre-training.
- **Agent**
  - Investigated a lightweight latent-space agent to integrate execution feedback and improve closed-loop decision-making efficiency in VLA/WAM.

Amazon | **Industrial Robotics Group (IRG)** | *Intern* | North Reading, MA *May. 2026 – Aug. 2026*

***Mentor:** Dr. Kai Gao **Manager:** Dr. Jeroen van Baar*

- **Long-horizon robot task with reasoning world model**
  - Developing world-model-based reasoning frameworks for long-horizon robotic manipulation, integrating visual evidence, future-state prediction, and VLA feedback to improve decision quality and task robustness.

Socure | **Computer Vision Team** | *Intern* | Remote *Jun. 2025 – Aug. 2025*

***Mentor:** Dr. Weqiang Mu **Manager:** Dr. Feng Xiao*

- **Low-Level Vision**
  - Developed a computational framework for bar-code reconstruction to mitigate occlusion, blur, and partial data loss, improving decoding robustness and accuracy.

AinnovationLabs | *Applied Scientist* | Remote *Dec. 2022 – Dec. 2024*

***Manager:** Prof. Yun Raymond Fu*

- **Healthcare AI**
  - Invented **1 patent** for sleep signal processing that identifies sleep stages and detects sleep disorders from complex acoustic signals.
  - Developed a mobile app to record sleep patterns and answer sleep-related questions, enabling large-scale

multimodal data collection for sleep research.

- **Computer Vision**
  - Developed a spatio-temporal 3D human pose estimation model with a self-collected dataset for accurate single-person motion tracking.

PUBLICATION

Under Review / Preprint

[**U-1**] ***Mingyuan Zhang***, Yue Bai, Huan Wang, Yizhou Wang, Qihua Dong, Yitian Zhang, and Yun Fu, "Boosting Large Language Models with Mask Fine-Tuning". [Paper] [Code]

[**U-2**] ***Mingyuan Zhang***, Yue Bai, Yifan Wang, Yiyang Huang, and Yun Fu, "Rethinking Fine-Tuning: Unlocking Hidden Capabilities in Vision-Language Models". [Paper]

[**U-3**] ***Mingyuan Zhang***, Yue Bai, Zhongruo Wang, Yupin Huang, Yiyang Huang, Hailing Wang, Huimin Zeng and Yun Fu, "MASKerade: Token-Routed Mask Experts for Dense-to-MoE Upcycling". [Paper]

[**U-4**] ***Mingyuan Zhang***, Jianglin Lu, Yiyang Huang, Yitian Zhang, Yizhou Wang and Yun Fu, "Selective Reduction in Foundation Models: A Survey".

[**U-5**] ***Mingyuan Zhang***, Ruining Yang, Yi Xu and Yun Fu, "Sports-Traj++: Unified, Disentangled, and Multi-Modal Trajectory Generation for Multi-Agent Movement in Sports".

[**U-6**] Yue Bai*, ***Mingyuan Zhang****, Handong Zhao, and Yun Fu, "Advancing Multi-Modal Foundation Models with Adapter Ensemble Strategies".

[**U-7**] Yi Xu, Ruining Yang, Yitian Zhang, Yizhou Wang, Jianglin Lu, ***Mingyuan Zhang***, Lili Su, and Yun Fu, "Trajectory Prediction Meets Large Language Models: A Survey". [Paper] [Code]

[**U-8**] Jianglin Lu, Hailing Wang, Xu Ma, Qihua Dong, ***Mingyuan Zhang***, Yizhou Wang, and Yun Fu, "MUSE: A Unified Multimodal Harness for MLLMs". [Paper] [Code]

[**U-9**] Yiyang Huang, Yitian Zhang, Yizhou Wang, Jianglin Lu, Qihua Dong, Hailing Wang, Huimin Zeng, ***Mingyuan Zhang*** and Yun Fu, "RACER: Reflective Agent Coupling Query Interpretation and Tool-Based Retrieval for Frame Selection in Long Video Understanding". [Paper]

Conference

[**C-1**] Linhan Wang, Zijian An, ***Mingyuan Zhang***, Chen Dai, Yi Xu, Can Cui, Zichong Yang, Yinlin Chen, Lifeng Zhou, Chang-Tien Lu, "GlanceWAM: Sparse Test-Time Imagination for World-Action Models", ***IROS 2026 RoBoWoMo Workshop (Spotlight & Best Award Honorable Mention)***. [Paper] [Code]

[**C-2**] Yiyang Huang, Yitian Zhang, Yizhou Wang, ***Mingyuan Zhang***, Liang Shi, Huimin Zeng, and Yun Fu, "Distorted or Fabricated? A Survey on Hallucination in Video LLMs", ***ACL 2026***. [Paper] [Code]

[**C-3**] Yizhou Wang, Lingzhi Zhang, Yue Bai, Mang Tik Chiu, Zhengmian Hu, ***Mingyuan Zhang***, Qihua Dong, Yu Yin, Sohrab Amirghodsi, and Yun Fu, "Cautious Next Token Prediction", ***ACL 2025***. [Paper] [Code]

Journal

[**J-1**] Yue Bai*, ***Mingyuan Zhang****, Huan Wang, Zhiqiang Tao, Kunpeng Li, and Yun Fu, "Random Sparse Networks Training with Sharpness-Aware Regularization", ***TPAMI***. [Paper]

PROJECTS

[**P-1**] Designed and developed a **mobile experimental logging system** for real-time data collection, enabling researchers to record structured experiment information on mobile devices and streamline the data-collection workflow.

[**P-2**] Designed and developed **Unity-based interactive games**, implementing gameplay mechanics, scene construction, user interaction, and narrative world-building.

ACADEMIC SERVICE

Reviewer:

- Conference: ACMMM 2024, AAAI 2025, UAI 2025, ACMMM 2025, NeurIPS 2025, ICRA 2027
- Journal: TKDD, TPAMI

Teaching:

- TA of EECE5639 Computer Vision, Spring 2026, NEU BOS